

Integral Recognition Guide (Exam-Oriented)

Which method should I use?

1. $f'(x)/f(x) \rightarrow \ln|f(x)|$
2. $f(x) \cdot f'(x) \rightarrow$ power rule after substitution
3. $(a + b \sin x)^n \cdot \cos x \, dx \rightarrow u = a + b \sin x$
4. $a^2 + x^2$ in denominator $\rightarrow \arctan$
5. $\sqrt{a^2 - x^2} \rightarrow \arcsin$ or $x = a \sin \theta$
6. $\sqrt{a^2 + x^2} \rightarrow \ln$ or $x = a \tan \theta$
7. Mixed $\sin$ & $\cos \rightarrow$ universal substitution $t = \tan(x/2)$
8. $\ln x$ with $1/x \rightarrow u = \ln x$
9. $e^{f(x)}$ with $f'(x) \rightarrow$ exponential substitution
10. None match $\rightarrow$ simplify algebraically first